

Improved Statistical Inference for Expectile-Based Risk Measures

An Optimal Pooling Approach in the Heavy-Tailed Regime

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Abstract

[**TODO:** Write a ~ 200 -word abstract once [Chapter 3](#) is in shape. Outline of the abstract: (i) expectiles as the only coherent and elicitable law-invariant risk measure; (ii) the problem of combining tail-index and extreme-quantile information from several samples (distributed inference, multi-study evidence synthesis); (iii) our pooled extreme-expectile estimator, its joint CLT via plug-in into the expectile–quantile asymptotic equivalence, and the variance-/AMSE-optimal weights tailored to the expectile target; (iv) Weissman-style extrapolation to very-extreme expectile levels; (v) outlook on relaxing the iid assumption.]

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Chapter 1

Introduction

1.1 Motivation

Quantitative risk management routinely needs to summarise the tail of a loss distribution by a single number. The industry-standard summary has long been the Value-at-Risk (VaR), defined as a high quantile of the loss distribution. VaR is intuitive and easy to compute, but it suffers from two well-known shortcomings that have shaped the search for alternatives. First, VaR is not subadditive: it can deem a diversified portfolio more risky than its components, contradicting the diversification principle and disqualifying VaR from the class of *coherent* risk measures in the sense of Artzner et al. (1999). Expected Shortfall (ES) corrects this defect and has progressively replaced VaR in regulatory practice. However, ES is not *elicitable* (Gneiting 2011): no scoring function has its expectation uniquely minimised at the true ES, so ES forecasts cannot be ranked or backtested directly.

Expectiles are the only law-invariant risk measure that is simultaneously coherent (for confidence levels $\tau \geq 1/2$) and elicitable (Bellini et al. 2014). Originally introduced by Newey and Powell (1987), the τ -expectile of a random loss X is defined as the minimiser of an asymmetric squared loss:

$$\xi_\tau(X) = \arg \min_{\theta \in \mathbb{R}} \mathbb{E} \left[|\tau - \mathbb{1}\{X \leq \theta\}| (X - \theta)^2 \right].$$

Setting $\tau = 1/2$ recovers the mean; setting $\tau \rightarrow 1$ moves the expectile into the right tail. Because the asymmetric loss is squared rather than absolute, expectiles — unlike quantiles — are sensitive to the *magnitude* of tail losses, not just to their frequency. Ehm et al. (2016) place expectiles within a broader theory of consistent scoring functions and Choquet representations.

This thesis addresses inference for expectiles at levels τ close to one, in two situations that are increasingly relevant in practice:

- (i) the loss distribution is heavy-tailed, in the sense of having an extreme-value index $\gamma > 0$, so that the empirical asymmetric-least-squares estimator becomes unstable as τ approaches one and extreme-value theory becomes the natural framework;

- (ii) the data are spread across several independent samples — whether because of distributed computation, regulatory data-sharing restrictions, or natural heterogeneity between studies — and inference must be conducted by communicating only summary statistics between sources.

1.2 Extreme expectile inference

The central bridge between expectile estimation and classical extreme-value theory is the *expectile-quantile asymptotic equivalence*. Under heavy-tailed assumptions with extreme-value index $\gamma \in (0, 1/2)$,

$$\frac{\xi_\tau(X)}{Q_X(\tau)} \xrightarrow{\tau \uparrow 1} \psi(\gamma) := (\gamma^{-1} - 1)^{-\gamma}, \quad (1.1)$$

where $Q_X(\tau)$ is the τ -quantile of X . Two families of estimators arise naturally from (1.1). The *asymmetric-least-squares* (LAWS) estimator minimises the empirical asymmetric criterion over the data; the *quantile-based* (QB) estimator substitutes consistent estimators of γ and of an extreme quantile of X into the asymptotic-equivalence formula. Their asymptotic theory at intermediate levels and via Weissman-type extrapolation to very-extreme levels was developed for iid heavy-tailed data by Daouia, Girard, et al. (2018) and subsequent work, and extended to the dependent setting by Davison et al. (2023); to the multivariate setting by Padoan and Stupfler (2022). The relevant machinery is reviewed in Chapter 2.

1.3 Pooled inference across samples

When m samples are available, with heterogeneous sizes and possibly heterogeneous tails, the natural question is how best to combine them. Daouia, Padoan, et al. (2021) develop an optimal pooling framework for the tail index γ and for extreme quantiles. Their estimators take the form of weighted arithmetic combinations (for the tail index) and weighted *geometric* combinations (for extreme quantiles), the latter specifically adapted to the multiplicative-power structure of the Weissman extrapolation. They derive the joint asymptotic normality of the per-sample estimators across m samples, allowing for asymptotic dependence between samples driven by a tail-copula, and identify the variance-optimal and AMSE-optimal pooling weights in closed form. In the important special case of *distributed inference* — where data are iid both within and across machines — the cross-sample tail-copula vanishes, off-diagonal entries of the joint asymptotic covariance collapse to zero, and the pooled estimators recover the asymptotic efficiency of the unfeasible Hill / Weissman estimator built from the combined sample.

The framework of Daouia, Padoan, et al. (2021) is the methodological foundation of this thesis. The framework does not address expectiles; the question of how to combine m pairs $(\hat{\gamma}_j, \hat{Q}_j(p_j))$ into a coherent inference procedure for an extreme expectile ξ_{τ_n} has not been studied.

1.4 Research question and scope

The thesis answers the following question:

Given m independent samples of iid heavy-tailed observations, with each sample j contributing point estimates $(\hat{\gamma}_j, \hat{Q}_j(p_j))$ together with their joint asymptotic covariance, how should one combine these summary statistics to perform inference and Weissman-style extrapolation for an extreme expectile of the common underlying distribution?

The scope is restricted as follows:

- the data within each sample are univariate iid (no time-series dependence), matching the distributed-inference setting of Daouia, Padoan, et al. (2021);
- the loss distribution is heavy-tailed with extreme-value index $\gamma \in (0, 1/2)$ — the standard range that guarantees the existence of the expectile and the validity of the asymptotic equivalence (1.1);
- the input to the pooling procedure is summary statistics, not raw data, in line with practical distributed-inference constraints.

1.5 Contributions

1. We define a *pooled extreme-expectile estimator* at intermediate levels by feeding the pooled tail-index estimator $\hat{\gamma}_n(\omega_\gamma)$ and the pooled weighted-geometric Weissman extreme-quantile estimator $\hat{q}_n^*(\cdot; \omega_q)$ of Daouia, Padoan, et al. (2021) into the asymptotic-equivalence formula (1.1).
2. We derive the joint asymptotic distribution of the pooled inputs, accounting for the cross-source structure (which is diagonal because the samples are independent) and for the within-source dependence between $\hat{\gamma}_j$ and $\hat{Q}_j$ induced by the use of overlapping top order statistics.
3. Applying the delta method through ψ , we obtain the asymptotic normality of the pooled intermediate-level expectile estimator together with explicit forms of its asymptotic bias and variance. The plug-in CLT of Davison et al. (2023, Proposition C.6) furnishes the key technical lemma.
4. We characterise the *variance-optimal* and *AMSE-optimal* pooling weights tailored to the pooled expectile estimator. These need not coincide with the weights optimal for the tail index or for the extreme quantile alone, owing to the non-trivial dependence between the two pieces of information.
5. We extend the construction to very-extreme levels $\tau'_n \uparrow 1$ with $n(1 - \tau'_n) \rightarrow c < \infty$ via a Weissman-style extrapolation of the pooled intermediate-level estimator, and derive the corresponding central limit theorem. As in the single-sample case, the extrapolated estimator inherits the asymptotic variance of the pooled tail-index estimator.
6. We construct asymptotic confidence intervals for extreme expectiles and a test for *expectile-tail-homogeneity* across the m samples, modelled on the tail-quantile-homogeneity test of

Daouia, Padoan, et al. (2021) and on the GLS-weighted equality-of-means test of Padoan and Stupfler (2022, Section 3.3).

1.6 Outline of the thesis

[Chapter 2](#) collects the technical background: regular variation and the second-order condition, the Hill and Weissman estimators, expectiles and their characterisation, single-sample iid extreme-expectile inference, and the optimal pooling framework of Daouia, Padoan, et al. (2021). [Chapter 3](#) develops the pooled estimator at intermediate expectile levels and the associated asymptotic theory. [Chapter 4](#) treats the Weissman-style extrapolation to very-extreme expectile levels. [Chapter 5](#) discusses limitations, the link to Expected Shortfall, and avenues for future work, including extensions to time-series dependence and multivariate observations. Detailed proofs are gathered in [Appendix A](#).

Chapter 2

Background

Outline of the chapter

This chapter collects the technical ingredients on which the thesis builds: regular variation and the second-order condition; the Hill and Weissman estimators for the tail index and extreme quantiles; expectiles and their characterisation as a coherent and elicitable risk measure; single-sample iid extreme-expectile inference; and the optimal pooling framework of Daouia, Padoan, et al. (2021). Nothing in this chapter is original; the contributions begin in [Chapter 3](#).

2.1 Regular variation and the second-order condition

Throughout the thesis we work with a real-valued random loss X that has continuous distribution function F and survival function $\bar{F} = 1 - F$. Given an iid sample $X_1, \dots, X_n$ from F , the corresponding order statistics are denoted by $X_{1:n} \leq X_{2:n} \leq \dots \leq X_{n:n}$.

Heavy-tailed asymptotics are governed by the language of regular variation; the reference is Haan and Ferreira (2006, Ch. 1).

Definition 2.1 (Regular variation). A measurable function $f : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is *regularly varying at infinity with index* $\alpha \in \mathbb{R}$, written $f \in \text{RV}_\alpha$, if for every $y > 0$,

$$\lim_{t \rightarrow \infty} \frac{f(ty)}{f(t)} = y^\alpha.$$

The functions in RV_0 are called *slowly varying*.

The first-order condition under which all of our asymptotics are taken asserts that F has a Pareto-type tail.

Condition 2.2 (First-order condition). There exists $\gamma > 0$ such that $\bar{F} \in \text{RV}_{-1/\gamma}$, or equivalently the *tail quantile function*

$$U(t) := \inf \left\{ x \in \mathbb{R} : \frac{1}{\bar{F}(x)} \geq t \right\}$$

satisfies $U \in \text{RV}_\gamma$.

Condition 2.2 is equivalent to saying that X lies in the max-domain of attraction of the Fréchet distribution with shape γ (Haan and Ferreira 2006, Theorem 1.2.1). The parameter $\gamma > 0$ is the *extreme-value index* of F ; it controls how slowly $\bar{F}$ decays at infinity. Standard heavy-tailed distributions encountered in risk management satisfy **Condition 2.2**, including the Pareto, Burr, Student- t , Fréchet, and F -distribution families (Haan and Ferreira 2006, §2.6).

For the asymptotic-normality statements that drive the inferential theory we need a quantitative refinement of **Condition 2.2** controlling the rate of convergence in **Definition 2.1**. The standard refinement is the second-order condition.

Condition 2.3 (Second-order condition $\mathcal{C}_2(\gamma, \rho, A)$). There exist $\gamma > 0$, a second-order parameter $\rho \leq 0$, and a measurable auxiliary function A of constant sign with $A(t) \rightarrow 0$ as $t \rightarrow \infty$, such that for every $y > 0$,

$$\lim_{t \rightarrow \infty} \frac{1}{A(t)} \left[\frac{U(ty)}{U(t)} - y^\gamma \right] = y^\gamma \frac{y^\rho - 1}{\rho},$$

where the right-hand side is read as $y^\gamma \log y$ when $\rho = 0$.

The function $|A|$ is itself regularly varying with index ρ (Haan and Ferreira 2006, Theorem 2.3.9); smaller $|\rho|$ means slower convergence and therefore larger second-order bias of the tail-index estimators we shall encounter in **Section 2.2**.

For the expectile–quantile asymptotic equivalence (**Section 2.4**) we will impose the stronger range $\gamma \in (0, 1/2)$, which guarantees both that X has finite second moment and that the expectile $\xi_\tau(X)$ admits a non-degenerate $\tau \uparrow 1$ limit when scaled by the corresponding quantile $Q_X(\tau)$.

2.2 Tail-index and extreme-quantile estimation

Under the first-order condition, the extreme-value index $\gamma > 0$ is the primary parameter determining the behaviour of F in the tail. We collect here the two classical estimators that the thesis builds on: the Hill estimator of γ and the Weissman extrapolation for extreme quantiles. Both are standard; the textbook reference is Haan and Ferreira (2006, Ch. 3–4).

2.2.1 The Hill estimator

Given an iid sample $X_1, \dots, X_n$ from F and an integer $k = k_n \in \{1, \dots, n-1\}$, the *Hill estimator* of γ based on the $k+1$ top order statistics is

$$\hat{\gamma}_n(k) = \frac{1}{k} \sum_{i=1}^k \log \left(\frac{X_{n-i+1:n}}{X_{n-k:n}} \right). \quad (2.1)$$

The interpretation is direct: if the upper tail of F were exactly Pareto with index $1/\gamma$, the log-spacings $\log(X_{n-i+1:n}) - \log(X_{n-k:n})$ would be iid exponential with mean γ , and $\hat{\gamma}_n(k)$ would be their sample mean. **Condition 2.2** asserts that the upper tail is *approximately* Pareto, which translates into the following asymptotic theory (Haan and Ferreira 2006, Theorem 3.2.5).

Theorem 2.4. Assume $\mathcal{C}_2(\gamma, \rho, A)$ with $\gamma > 0$. Let $k = k_n \rightarrow \infty$ and $k/n \rightarrow 0$ as $n \rightarrow \infty$. Then

$\hat{\gamma}_n(k) \xrightarrow{\mathbb{P}} \gamma$. If, in addition, $\sqrt{k} A(n/k) \rightarrow \lambda \in \mathbb{R}$, then

$$\sqrt{k} (\hat{\gamma}_n(k) - \gamma) \xrightarrow{d} \mathcal{N}\left(\frac{\lambda}{1-\rho}, \gamma^2\right).$$

The choice of k controls the well-known bias–variance trade-off: too small a k inflates the variance, while too large a k injects second-order bias through the term $\lambda/(1-\rho)$. Adaptive choice of k is a research topic in its own right; we do not pursue it in this thesis and treat k as fixed by the data analyst.

2.2.2 Weissman extrapolation to extreme quantiles

The *Weissman extrapolation* (Weissman 1978) extrapolates from the intermediate-order statistic $X_{n-k:n}$ — which estimates the quantile $Q_X(1-k/n)$ at the intermediate level $1-k/n$ — to a quantile at a very-extreme level $1-p$ with $p = p_n \rightarrow 0$ as $n \rightarrow \infty$. Plugging the Hill estimator into the first-order identity $U(ty)/U(t) \rightarrow y^\gamma$ gives the Weissman estimator

$$\hat{q}_n^*(1-p | k) = \left(\frac{k}{np}\right)^{\hat{\gamma}_n(k)} X_{n-k:n}. \quad (2.2)$$

The typical regime of interest is np bounded, i.e. extrapolation beyond the largest observation in the sample.

Theorem 2.5. Assume $\mathcal{C}_2(\gamma, \rho, A)$ with $\gamma > 0$ and $\rho < 0$. Let $k = k_n \rightarrow \infty$, $k/n \rightarrow 0$, $p = p_n \rightarrow 0$ with $k/(np) \rightarrow \infty$ and $\sqrt{k}/\log(k/(np)) \rightarrow \infty$. Suppose $\sqrt{k} A(n/k) \rightarrow \lambda \in \mathbb{R}$. Then

$$\frac{\sqrt{k}}{\log(k/(np))} \left(\frac{\hat{q}_n^*(1-p | k)}{Q_X(1-p)} - 1 \right) \xrightarrow{d} \mathcal{N}\left(\frac{\lambda}{1-\rho}, \gamma^2\right).$$

The rate $\sqrt{k}/\log(k/(np))$ reflects the cost of extrapolation: the further into the tail one extrapolates, the larger the log-denominator and the slower the rate. The asymptotic variance is identical to that of the Hill estimator (Theorem 2.4); the extrapolation contributes only through the multiplicative log-scale factor. This is the structural fact that Daouia, Padoan, et al. (2021, §2.3) exploits to motivate the weighted-*geometric* pooling of Weissman estimators — rather than the arithmetic pooling natural for centred quantities — and that we shall in turn exploit in Chapter 4 for the extrapolation of pooled expectiles.

2.3 Expectiles

[**TODO:** Definition via the asymmetric-LS minimisation of Newey and Powell (1987); first-order conditions for the population expectile; basic properties (existence for $\mathbb{E}|X| < \infty$, uniqueness, location/scale equivariance, monotonicity in τ , $\xi_{1/2} = \mathbb{E}X$). Coherence (for $\tau \geq 1/2$) and elicibility via the asymmetric squared scoring function; cite Bellini et al. (2014). Choquet representation of Ehm et al. (2016). Risk-management interpretation as a VaR-distortion.]

2.4 Extreme expectile inference: the iid heavy-tailed baseline

[**TODO:** State and reference the expectile–quantile asymptotic equivalence: $\xi_\tau/Q_X(\tau) \rightarrow (\gamma^{-1} - 1)^{-\gamma}$ as $\tau \uparrow 1$ for $\gamma \in (0, 1)$ (Daouia, Girard, et al. 2019, Proposition 1).]

[**TODO:** The two estimator families in the iid setting (Daouia, Girard, et al. 2018): (a) LAWS direct, $\tilde{\xi}_{\tau_n} = \arg \min_{\theta} \sum_i \eta_{\tau_n}(X_i - \theta)$, with asymptotic variance $2\gamma^3/(1 - 2\gamma)$ at intermediate levels; (b) QB plug-in, $\hat{\xi}_{\tau_n} = (\hat{\gamma}^{-1} - 1)^{-\hat{\gamma}} \hat{Q}(\tau_n)$, with asymptotic variance involving $m(\gamma) = (1 - \gamma)^{-1} - \log(\gamma^{-1} - 1)$.]

[**TODO:** Weissman-style extrapolation to very-extreme levels τ'_n with $n(1 - \tau'_n) \rightarrow c$: $\bar{\xi}_{\tau'_n}^* = ((1 - \tau'_n)/(1 - \tau_n))^{-\hat{\gamma}} \bar{\xi}_{\tau_n}$, with asymptotic variance γ^2 in the iid case. The LAWS-anchored and QB-anchored extrapolations have the same asymptotic distribution at the extreme level (the tail-index estimator dominates).]

[**TODO:** State the plug-in CLT of Davison et al. (2023, Proposition C.6), which is the abstract technical lemma that lifts joint $\sqrt{n(1 - \tau_n)}$ -Gaussian limits of $(\hat{\gamma}, \bar{\xi})$ to limits of the extrapolated $\bar{\xi}^*$. This is the workhorse for Chapter 3 and Chapter 4.]

2.5 The optimal pooling framework

[**TODO:** Setup: m samples $\mathbf{X}_1, \dots, \mathbf{X}_m$, sample sizes n_j , total $n = \sum n_j$; effective sample sizes k_j for tail estimation; ratios $n_j/n \rightarrow b_j$, $k_j/n_j \rightarrow 0$, $k_j/k \rightarrow c_j$. Marginal second-order condition $\mathcal{C}_2(\gamma_j, \rho_j, A_j)$. Tail-copula assumption $\mathcal{J}(\mathbf{R})$ encoding cross-sample asymptotic dependence via $R_{j,\ell}$. The common-tail case $\gamma = \gamma \mathbf{1}$ that the thesis adopts.]

[**TODO:** Pooled Hill estimator $\hat{\gamma}_n(\boldsymbol{\omega}) = \boldsymbol{\omega}^\top \hat{\gamma}_n$ with $\boldsymbol{\omega}^\top \mathbf{1} = 1$. State Theorem 1 of Daouia, Padoan, et al. (2021): joint $\sqrt{k_j}(\hat{\gamma}_j(k_j) - \gamma_j) \xrightarrow{d} \mathcal{N}(\mathbf{B}, \mathbf{V})$, with bias vector $\mathbf{B}$ involving $\lambda_j/(1 - \rho_j)$ and covariance matrix $\mathbf{V}$ whose diagonal entries are γ_j^2 and off-diagonal entries are $\gamma_j \gamma_\ell \sqrt{c_\ell/c_j} R_{j,\ell}(c_\ell/c_j, b_\ell/b_j)$.]

[**TODO:** Variance-optimal and AMSE-optimal weights: $\boldsymbol{\omega}^{\text{Var}} = \mathbf{V}_c^{-1} \mathbf{1}/(\mathbf{1}^\top \mathbf{V}_c^{-1} \mathbf{1})$, and the AMSE-optimal counterpart involving $\mathbf{B}_c$ and $\mathbf{V}_c$.]

[**TODO:** Weighted geometric pooled Weissman estimator $\hat{q}_n^*(1 - p | \boldsymbol{\omega}) = \prod_j [\hat{q}_j^*(1 - p | k_j)]^{\omega_j}$ and Theorem 2 of Daouia, Padoan, et al. (2021) giving its asymptotic distribution.]

[**TODO:** Distributed-inference specialisation: iid data within and across machines forces $R_{j,\ell} \equiv 0$, so $\mathbf{V}$ becomes diagonal with entries γ^2/c_j ; the variance-optimal weights reduce to $\omega_j^{\text{Var}} \propto k_j$ (effective-sample-size weighting); and the pooled Hill and Weissman estimators are asymptotically equivalent to the unfeasible benchmark Hill and Weissman estimators built from the combined sample.]

[**TODO:** Tail-homogeneity test $H_0 : \gamma = \gamma \mathbf{1}$ via the chi-square statistic $\Lambda_n = k(\hat{\gamma}_n - \bar{\mu}_n \mathbf{1})^\top \widehat{\mathbf{V}}_c^{-1}(\hat{\gamma}_n - \bar{\mu}_n \mathbf{1})$. Asymptotic confidence interval for γ under tail homogeneity.]

Chapter 3

Pooled inference for extreme expectiles at intermediate levels

Overview

This chapter introduces the pooled extreme-expectile estimator at intermediate levels and develops its asymptotic theory. The construction plugs the pooled tail-index estimator $\hat{\gamma}_n(\boldsymbol{\omega}_\gamma)$ and the pooled weighted-geometric Weissman extreme-quantile estimator $\hat{q}_n^*(\cdot; \boldsymbol{\omega}_q)$ of Daouia, Padoan, et al. (2021) into the expectile–quantile asymptotic equivalence $\xi_\tau \approx \psi(\gamma) Q_X(\tau)$. The delta method applied through ψ yields a joint Gaussian limit, from which we read off the variance- and AMSE-optimal pooling weights tailored to the pooled expectile target.

3.1 Setup and notation

[**TODO:** We have m independent samples of iid heavy-tailed observations with a common extreme-value index $\gamma \in (0, 1/2)$. Each sample j contributes a Hill estimator $\hat{\gamma}_j(k_j)$ and a Weissman extreme-quantile estimator $\hat{q}_j^*(1 - p \mid k_j)$, together with their joint asymptotic covariance in the sense of Theorems 1 and 2 of Daouia, Padoan, et al. (2021). Because the samples are *independent*, the tail-copula functions $R_{j,\ell}$ are identically zero across sources, and the cross-source blocks of the joint asymptotic covariance vanish. We target the extreme expectile ξ_{τ_n} at intermediate level $\tau_n \rightarrow 1$ with $n(1 - \tau_n) \rightarrow \infty$.]

3.2 The pooled QB expectile estimator

[**TODO:** Define the pooled QB extreme-expectile estimator

$$\hat{\xi}_{\tau_n}^{\text{pool}}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) := \psi(\hat{\gamma}_n(\boldsymbol{\omega}_\gamma)) \hat{q}_n^*(1 - \tau_n \mid \boldsymbol{\omega}_q), \quad \psi(g) = (g^{-1} - 1)^{-g}.$$

Discuss the choice of weight vectors $\boldsymbol{\omega}_\gamma$ and $\boldsymbol{\omega}_q$ — they need not coincide — and motivate the use of the *intermediate-level* pooled Weissman (rather than the empirical intermediate quantile)

by appealing to the asymptotic equivalence between the two at intermediate levels.]

3.3 Joint asymptotic distribution of the pooled inputs

[**TODO:** State and prove the joint CLT for the $2m$ -vector $(\hat{\gamma}_j(k_j), \hat{q}_j^*(1 - \tau_n | k_j))_{j=1, \dots, m}$ under the iid baseline. The within-sample 2×2 blocks coincide with the joint $(\hat{\gamma}_j, \hat{q}_j)$ limit of Padoan and Stupfler (2022, Theorem 2.2 and Corollary 2.3); the between-sample blocks vanish because the samples are independent. Conclude that the pooled $(\hat{\gamma}_n(\omega_\gamma), \hat{q}_n^*(1 - \tau_n | \omega_q))$ is jointly Gaussian with covariance

$$\Sigma(\omega_\gamma, \omega_q) = \begin{pmatrix} \omega_\gamma^\top \mathbf{V}_c \omega_\gamma & \text{cross-covariance} \\ \cdot & \omega_q^\top \mathbf{V}_c \omega_q \end{pmatrix},$$

with the cross-covariance evaluated explicitly from the joint within-sample distribution of $(\hat{\gamma}_j, \hat{q}_j)$.]

3.4 Asymptotic normality of the pooled expectile estimator

[**TODO:** Apply the delta method to the joint Gaussian limit of the previous section through the map $(g, q) \mapsto \psi(g)q$. The Jacobian is $(\psi'(g)q, \psi(g))$ with $\psi'(g) = \psi(g) \cdot [(1 - g)^{-1} - \log(g^{-1} - 1)]/g$, which factor we denote $m(\gamma)$. The resulting limit is

$$\sqrt{k} \left(\frac{\hat{\xi}_{\tau_n}^{\text{pool}}}{\xi_{\tau_n}} - 1 \right) \xrightarrow{d} \mathcal{N}(\mu^\xi(\omega_\gamma, \omega_q), \sigma^\xi(\omega_\gamma, \omega_q)^2),$$

with explicit forms of the asymptotic bias μ^ξ and variance $\sigma^{\xi,2}$ in terms of $\mathbf{B}_c$ and $\mathbf{V}_c$. Cite the *plug-in CLT of Davison et al. (2023, Proposition C.6)* as the abstract technical lemma justifying this manipulation.]

3.5 Variance- and AMSE-optimal weights

[**TODO:** Derive closed-form expressions for the weights minimising $\sigma^{\xi,2}$ and the AMSE $(\mu^\xi)^2 + \sigma^{\xi,2}$, respectively. The minimisation is over $(\omega_\gamma, \omega_q)$ subject to $\omega_\gamma^\top \mathbf{1} = \omega_q^\top \mathbf{1} = 1$. Show that the optimal weights differ from ω^{Var} of Daouia, Padoan, et al. (2021) whenever the within-sample cross-covariance between $\hat{\gamma}_j$ and $\hat{q}_j$ is non-zero, and identify the explicit difference. Provide a sufficient condition for the optimal expectile weights to coincide with the optimal quantile weights, namely $m(\gamma) \rightarrow 0$ (i.e. γ near the value at which the LAWS and QB intermediate variances coincide).]

3.6 Estimated weights and confidence intervals

[**TODO:** Plug-in estimation of $\mathbf{B}_c$ and $\mathbf{V}_c$ from the per-sample $(\hat{\gamma}_j, \hat{q}_j)$ pairs, following Section 2.2 of Daouia, Padoan, et al. (2021). Asymptotic $(1 - \alpha)$ -confidence interval for ξ_{τ_n} built on the Gaussian limit of the previous section.]

3.7 A test of expectile-tail-homogeneity

[**TODO:** Consider the null hypothesis that the m samples share a common expectile-tail behaviour, i.e. $\xi_{\tau,j}/\xi_{\tau,\ell} \rightarrow 1$ for all pairs (j, ℓ) , which under tail homogeneity is equivalent to $\gamma_j = \gamma$ for all j and $q_{\tau,j}/q_{\tau,\ell} \rightarrow 1$ (Daouia, Padoan, et al. 2021, Remark 5). Following the GLS construction of Padoan and Stupfler (2022, §3.3), the χ^2_{m-1} deviance statistic on the log-scale yields a test of size α for expectile-tail-homogeneity. Reject for large values and accept otherwise.]

Chapter 4

Extrapolation to very-extreme expectile levels

Overview

This chapter extrapolates the pooled intermediate-level expectile estimator of [Chapter 3](#) to very-extreme levels $\tau'_n \uparrow 1$ with $n(1 - \tau'_n) \rightarrow c \in [0, \infty)$. The extrapolation is the analogue of the Weissman extrapolation for extreme quantiles (Weissman [1978](#)) and follows the same multiplicative-power structure that motivates the weighted-geometric pooling of Daouia, Padoan, et al. ([2021](#), §2.3). As in the single-sample case (Davison et al. [2023](#), Theorem 3.5), the asymptotic distribution at the extreme level is dominated by the tail-index estimator: the contribution of the intermediate-level expectile estimator washes out through the log-rate.

4.1 Motivation

[**TODO:** Risk-management applications routinely require expectiles at levels τ'_n so close to one that there are essentially no observations in the corresponding tail region (typical regime: $n(1 - \tau'_n) \rightarrow c \in [0, \infty)$, equivalently $1 - \tau'_n \leq 1/n$). Direct estimation by either LAWS or QB at τ'_n is not feasible; one must extrapolate from an intermediate-level estimator using a model for the tail.]

4.2 The pooled extrapolated expectile estimator

[**TODO:** Define the pooled extrapolated estimator by Weissman-style lifting of the pooled intermediate-level estimator of [Chapter 3](#):

$$\hat{\xi}_{\tau'_n}^{\text{pool},*} := \left(\frac{1 - \tau'_n}{1 - \tau_n} \right)^{-\hat{\gamma}_n(\omega_\gamma)} \hat{\xi}_{\tau_n}^{\text{pool}}(\omega_\gamma, \omega_q).$$

Discuss two natural alternatives that turn out to be asymptotically equivalent: (i) plug the pooled *extrapolated* quantile $\hat{q}_n^*(1 - (1 - \tau'_n)/(\gamma^{-1} - 1) \mid \omega_q)$ into $\psi(\hat{\gamma}_n)$; (ii) the geometric pooling

of marginal extrapolated expectiles. Show via the multiplicative-power identity that all three constructions agree at the asymptotic level.]

4.3 Asymptotic distribution

[**TODO:** State and prove the central limit theorem

$$\frac{\sqrt{k}}{\log((1 - \tau_n)/(1 - \tau'_n))} \left(\frac{\hat{\xi}_{\tau'_n}^{\text{pool},*}}{\xi_{\tau'_n}} - 1 \right) \xrightarrow{d} \mathcal{N}\left(\mu^{\xi,*}(\omega_\gamma), \sigma^{\xi,*}(\omega_\gamma)^2\right),$$

where the limit law depends *only* on the pooled tail-index estimator (the contribution of $\hat{\xi}_{\tau_n}^{\text{pool}}$ enters through a log-multiplicative term that vanishes at the displayed rate). Explicitly, $\sigma^{\xi,*}(\omega_\gamma)^2 = \omega_\gamma^\top \mathbf{V}_c \omega_\gamma$. Cite Davison et al. (2023, Proposition C.6) as the plug-in lemma and follow the proof template of Davison et al. (2023, Theorem 3.5).]

4.4 Optimal weights for extrapolation

[**TODO:** Because the asymptotic variance is governed solely by $\hat{\gamma}_n(\omega_\gamma)$, the variance-optimal weights for extrapolation coincide with ω^{Var} of Daouia, Padoan, et al. (2021): $\omega^{\text{Var},*} = \mathbf{V}_c^{-1} \mathbf{1}/(\mathbf{1}^\top \mathbf{V}_c^{-1} \mathbf{1})$. In the distributed-inference case ($R_{j,\ell} \equiv 0$), these reduce to $\omega_j \propto k_j$ (effective-sample-size weighting), which is the same conclusion as for extreme-quantile extrapolation. Independently, ω_q becomes irrelevant at the extreme level.]

4.5 Bias-reduction

[**TODO:** Bias-reduced versions of the pooled tail index lift to bias-reduced versions of the pooled extrapolated expectile. Following the bias-reduction scheme of Daouia, Padoan, et al. (2021, §2.2), define $\bar{\gamma}_n(\omega) = \hat{\gamma}_n(\omega) - k^{-1/2} \omega^\top \widehat{\mathbf{B}}_c$, and propagate this into $\hat{\xi}_{\tau'_n}^{\text{pool},*}$. State the resulting unbiased CLT.]

4.6 Asymptotic confidence intervals

[**TODO:** Log-scale confidence intervals for $\xi_{\tau'_n}$ in the spirit of Daouia, Padoan, et al. (2021, Corollary 4):

$$\mathbb{P} \left(\xi_{\tau'_n} \in \hat{\xi}_{\tau'_n}^{\text{pool},*} \exp \left[\pm z_{1-\alpha/2} \log \left(\frac{1 - \tau_n}{1 - \tau'_n} \right) \sqrt{\frac{\omega_\gamma^\top \widehat{\mathbf{V}}_c \omega_\gamma}{k}} \right] \right) \rightarrow 1 - \alpha.$$

Discuss finite-sample performance heuristically.]

Chapter 5

Discussion and future work

5.1 Summary of contributions

[**TODO:** Restate the contributions of Chapters 3 and 4 in the language of Section 1.5. Emphasise the modular architecture: the pooled inputs $(\hat{\gamma}_n, \hat{q}_n^*)$ of Daouia, Padoan, et al. (2021) compose cleanly with the expectile–quantile asymptotic equivalence via the plug-in CLT of Davison et al. (2023, Prop. C.6), so the only expectile-specific computation is the delta-method calculus around $\psi(\gamma) = (\gamma^{-1} - 1)^{-\gamma}$.]

5.2 Connections to Expected Shortfall

[**TODO:** For a heavy-tailed loss with index $\gamma \in (0, 1)$ there is an asymptotic relation $\xi_\tau \sim \text{ES}_{\tau'}$ for a specific matching level $\tau' = \tau'(\tau, \gamma)$ that can be expressed in closed form (cf. Daouia, Girard, et al. (2018)). Thus pooled expectile inference yields, as an immediate corollary, pooled inference on the Expected Shortfall, sidestepping the non-elicitability of ES. Sketch the level-matching identity and the corresponding plug-in estimator.]

5.3 Limitations

[**TODO:** (i) Tail-homogeneity across samples is assumed when pooling extreme quantiles geometrically (cf. assumption $(\mathcal{H})$ in Daouia, Padoan, et al. (2021, §2.3)) and is also needed for pooling expectiles. The test of Section 3.7 should be applied first.

(ii) The choice of effective sample sizes k_j is data-driven and introduces variability that the asymptotic theory does not account for.

(iii) The asymptotic equivalence (1.1) requires $\gamma < 1/2$ for a finite second moment; very heavy tails ($\gamma \geq 1/2$) fall outside the scope.

(iv) The framework is iid within and across samples. Time-series dependence within a sample and dependence across samples are excluded.]

5.4 Future work

[**TODO:** (i) *Time-series dependence within each sample.* Extending the framework to β -mixing observations within a sample, in the spirit of Davison et al. (2023), would require inflating the within-sample asymptotic variance by a serial-dependence factor. The pooled inference architecture itself is preserved.

(ii) *Multivariate observations.* Each sample could be d -dimensional, with the goal of pooling marginal-component-wise expectiles. The joint expectile machinery of Padoan and Stupfler (2022) provides the within-sample tools; the cross-sample step is identical to the univariate case.

(iii) *Tail-heteroscedasticity.* A simple parametric model for $q_j(\tau)/q_\ell(\tau) \rightarrow \kappa_{j,\ell}$ would allow pooling even when assumption $(\mathcal{H})$ fails.

(iv) *Empirical applications.* Distributed-inference settings in operational risk (cross-bank loss data), insurance (multi-line claims), and climate (multi-station extremes) are natural use cases.

(v) *Bias-robust pooled estimators.* Use bias-reduced tail-index estimators (e.g. corrected Hill) in place of the plain Hill estimator in the pooled construction.]

Appendix A

Proofs

[**TODO:** Detailed proofs of the main theorems of [Chapters 3](#) and [4](#) go here. Recommended organisation: (A.1) joint CLT for the pooled $(\hat{\gamma}_n, \hat{q}_n^*)$ vector, (A.2) delta-method computation for the QB pooled expectile, (A.3) Weissman extrapolation lemma, (A.4) bias-component estimation.]

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